

Problem 29.9: Chiral Rotations and the Theta Angle

Source: M. D. Schwartz, *Quantum Field Theory and the Standard Model*, Problem 29.9.

Show that when multiple generations are rotated, then the θ angle shifts by $\arg \det(R^\dagger L)$.

Solution. For N generations, the chiral rotations are

$$\psi_R^i \rightarrow R^{ij} \psi_R^j, \quad \psi_L^i \rightarrow L^{ij} \psi_L^j, \quad (1.1)$$

where $R, L \in U(N)$. We decompose

$$U(N) = SU(N) \times U(1), \quad (1.2)$$

and write

$$R = e^{ir/N} \tilde{R}, \quad L = e^{i\ell/N} \tilde{L}, \quad (1.3)$$

where

$$\tilde{R}, \tilde{L} \in SU(N), \quad \det \tilde{R} = \det \tilde{L} = 1. \quad (1.4)$$

Under the $SU(N)$ transformations, the Grassmann measures transform as

$$\prod_i \mathcal{D}\psi_R^i \rightarrow (\det \tilde{R})^{-1} \prod_i \mathcal{D}\psi_R^i = \prod_i \mathcal{D}\psi_R^i \quad (1.5)$$

and

$$\prod_i \mathcal{D}\psi_L^i \rightarrow (\det \tilde{L})^{-1} \prod_i \mathcal{D}\psi_L^i = \prod_i \mathcal{D}\psi_L^i. \quad (1.6)$$

The conjugate-field measures are also invariant. Therefore, the $SU(N)$ parts of R and L do not change the complete functional measure. Only the $U(1)$ parts can shift the θ angle.

The remaining $U(1)$ transformations are

$$\psi_R \rightarrow e^{ir/N} \psi_R, \quad \psi_L \rightarrow e^{i\ell/N} \psi_L. \quad (1.7)$$

These can be written as a vector rotation and an axial rotation:

$$\psi \rightarrow e^{i\alpha_V} e^{i\alpha_A \gamma^5} \psi, \quad (1.8)$$

where

$$\alpha_V = \frac{r + \ell}{2N}, \quad \alpha_A = \frac{r - \ell}{2N}. \quad (1.9)$$

The vector rotation leaves the functional measure invariant. An axial rotation by α_A shifts the θ angle by $-2\alpha_A$ for each generation. Since there are N generations,

$$\Delta\theta = -2N\alpha_A = -2N \left(\frac{r - \ell}{2N} \right) = \ell - r. \quad (1.10)$$

Finally,

$$\begin{aligned} \det(R^\dagger L) &= (\det R)^* \det L \\ &= e^{-ir} e^{i\ell} \\ &= e^{i(\ell - r)}. \end{aligned} \quad (1.11)$$

Therefore,

$$\arg \det(R^\dagger L) = \ell - r = \Delta\theta, \quad (1.12)$$

and hence

$$\boxed{\theta \rightarrow \theta + \arg \det(R^\dagger L)}. \quad (1.13)$$